

Single-molecule enzyme-linked immunosorbent assay detects serum proteins at subfemtomolar concentrations

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The ability to detect single protein molecules^{1,2} in blood could accelerate the discovery and use of more sensitive diagnostic biomarkers. To detect low-abundance proteins in blood, we captured them on microscopic beads decorated with specific antibodies and then labeled the immunocomplexes (one or zero labeled target protein molecules per bead) with an enzymatic reporter capable of generating a fluorescent product. After isolating the beads in 50-fl reaction chambers designed to hold only a single bead, we used fluorescence imaging to detect single protein molecules. Our single-molecule enzyme-linked immunosorbent assay (digital ELISA) approach detected as few as ~10–20 enzyme-labeled complexes in 100 μ l of sample (~10⁻¹⁹ M) and routinely allowed detection of clinically relevant proteins in serum at concentrations (<10⁻¹⁵ M) much lower than conventional ELISA^{3–5}. Digital ELISA detected prostate-specific antigen (PSA) in sera from patients who had undergone radical prostatectomy at concentrations as low as 14 fg/ml (0.4 fM).

The clinical use of protein biomarkers to differentiate between healthy and disease states, and to monitor disease progression, requires the measurement of low concentrations of proteins in complex samples. Current immunoassays typically measure proteins at concentrations above 10⁻¹² M⁶. The serum concentrations of the majority of proteins important in cancer⁷, neurological disorders^{8,9} and the early stages of infection¹⁰, however, are thought to range from 10⁻¹⁶ to 10⁻¹² M. For instance, a 1-mm³ tumor composed of a million cells that each secrete 5,000 proteins into 5 liters of circulating blood translates to a concentration of ~2 $\times$ 10⁻¹⁵ M (or 2 fM). Moreover, serum from individuals recently infected with HIV contains 10–3,000 virions per ml, resulting in estimated concentrations of the p24 capsid antigen ranging from 50 $\times$ 10⁻¹⁸ M (50 aM) to 15 $\times$ 10⁻¹⁵ M (15 fM)¹⁰. Attempts to develop methods capable of measuring these concentrations of proteins have focused on the replication of nucleic acid labels on proteins^{11,12}, or on measuring the bulk, ensemble properties of labeled protein molecules^{13–16}. The work of Mirkin *et al.*^{12,17} and others¹⁸ using labels based on gold nanoparticles and DNA barcodes has pushed the detection of proteins into the low femtomolar range; a recent report

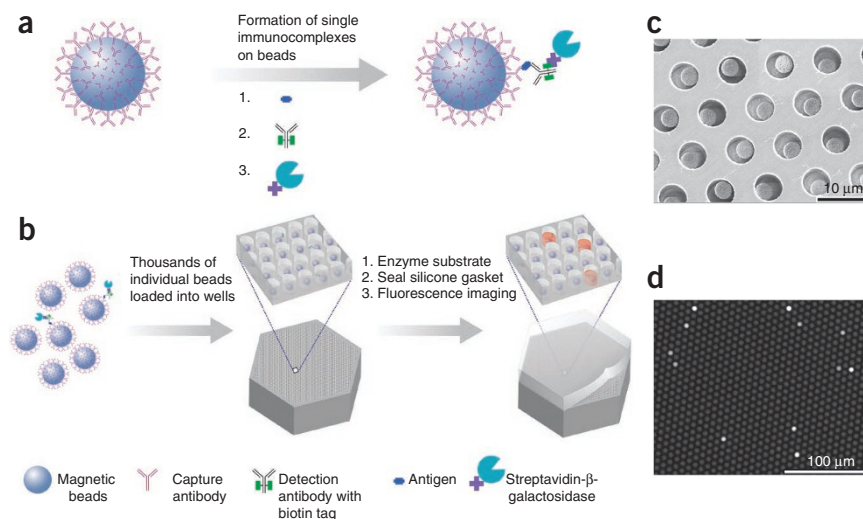
using this technology demonstrated the detection of 10 fM of PSA in serum¹⁷. Nonetheless, the sensitivities achieved by methods for detecting proteins still lag behind those for nucleic acids, such as PCR, limiting the number of gene products that have been detected in blood^{6,19}. The isolation and detection of single protein molecules provides a promising approach for measuring extremely low concentrations of proteins^{1,2}. For example, Todd *et al.*² have developed flow-based methods for serially detecting single fluorescently labeled detection antibodies that have been released from immunocomplexes formed on solid substrates. Here we report an approach for detecting thousands of single protein molecules simultaneously using the same reagents as the gold standard for detecting proteins, namely, the ELISA. This method has been used to detect proteins in serum at subfemtomolar concentrations.

Our approach makes use of arrays of femtoliter-sized reaction chambers (Fig. 1)—which we term single-molecule arrays (SiMoAs)—that can isolate and detect single enzyme molecules^{20–24}. This approach builds from the work of Walt *et al.*^{20–23}, who used these arrays to study the kinetics²¹ and inhibition²⁰ of single enzymes. Our objective was to exploit the ability of SiMoAs to trap and detect single enzymes to detect single enzyme-labeled proteins. In the first step of this single-molecule immunoassay (Fig. 1a), a sandwich antibody complex is formed on microscopic beads (2.7 μ m diameter), and the bound complexes are labeled with an enzyme, as in a conventional bead-based ELISA. When assaying samples containing extremely low concentrations of protein, the ratio of protein molecules (and the resulting enzyme-labeled complex) to beads is small (typically <1:1) and, as such, the percentage of beads that contain a labeled immunocomplex follows a Poisson distribution. At low concentrations of protein, the Poisson distribution indicates that beads carry either a single immunocomplex or none. For example, if 50 aM of a protein in 0.1 ml (3,000 molecules) is captured and labeled on 200,000 beads, then 1.5% of the beads will carry one protein molecule and 98.5% will not carry any protein molecules (Fig. 1b)²². It is not possible to detect these low numbers of enzyme labels using standard detection technology (for example, a plate reader), because the fluorophores generated by each enzyme diffuse into a large assay volume (typically 0.1–1 ml), and it takes hundreds of thousands of enzyme labels to generate a

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Figure 1 Digital ELISA based on arrays of femtoliter-sized wells. (a,b) Single protein molecules are captured and labeled on beads using standard ELISA reagents (a), and beads with or without a labeled immunoconjugate are loaded into femtoliter-volume well arrays for isolation and detection of single molecules by fluorescence imaging (b). (c) Scanning electron micrograph of a small section of a femtoliter-volume well array after bead loading. Beads (2.7 μm diameter) were loaded into an array of wells with diameters of 4.5 μm and depths of 3.25 μm . (d) Fluorescence image of a small section of the femtoliter-volume well array after signals from single enzymes are generated. Whereas the majority of femtoliter-volume chambers contain a bead from the assay, only a fraction of those beads possess catalytic enzyme activity, indicating a single, bound protein molecule. The concentration of protein in bulk solution is correlated to the percentage of beads that carry a protein molecule.



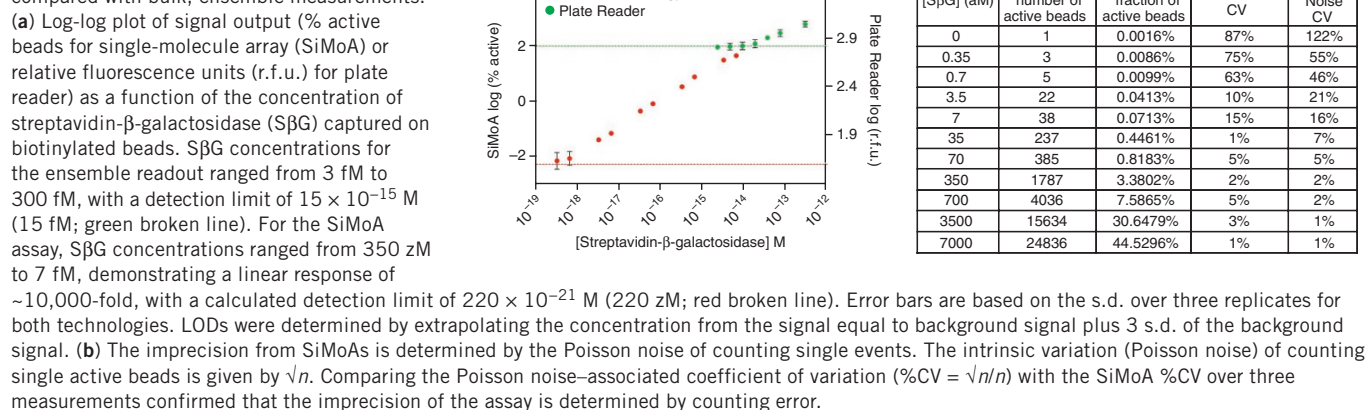
fluorescence signal above background. In contrast, SiMoAs permit the detection of very low concentrations of enzyme labels by confining the fluorophores generated by individual enzymes to extremely small volumes (~ 50 fl), ensuring a high local concentration of fluorescent product molecules. To achieve this confinement in our assay, beads are loaded into an array of femtoliter-sized wells; we used 2-mm-wide arrays with $\sim 50,000$ wells, each with a diameter of 4.5 μm and a depth of 3.25 μm (Fig. 1c). After sealing the loaded arrays against a rubber gasket in the presence of a droplet of fluorogenic enzyme substrate, each bead is isolated in a femtoliter-volume reaction chamber. Beads possessing a single enzyme-labeled immunocomplex generate a high concentration of fluorescent product that is restricted to the 50-fl reaction chamber (Fig. 1d). By acquiring time-lapsed fluorescence images of the array using standard microscope optics, it is possible to distinguish beads associated with a single enzyme molecule ("on" well) from those not associated with an enzyme ("off" well); **Supplementary Figure 1** shows histograms of fluorescence from "on" and "off" wells. Imaging the arrays allows simultaneous detection of tens to tens of thousands of single immunocomplexes. The protein concentration in the test sample is determined by counting the number of wells containing both a bead and fluorescent product relative to the total number of wells containing beads (Fig. 1d). As SiMoAs enable concentration to be determined digitally rather than by using the total analog signal, we call our approach to detecting single immunocomplexes digital ELISA.

We first assessed the intrinsic sensitivity of this strategy by creating populations of beads with well-characterized enzyme-to-bead ratios. We mixed 400,000 biotin-modified beads with a range of concentrations of the enzyme conjugate streptavidin- β -galactosidase (β Gal). For convenience, biotinylated beads were provided by hybridizing biotinylated DNA with beads functionalized with complementary DNA. (We note that this experiment should not be construed as a sensitive DNA assay; the sensitivity of such an assay is limited by nonspecific interactions between the enzyme conjugate and surface-bound DNA as shown in **Supplementary Fig. 2**.) These beads were detected in two different ways. First, we assayed an ensemble of beads in 100 μl using a fluorescence plate reader after 1 h incubation with 100 μM resorufin- β -D-galactopyranoside (RGP), a fluorogenic substrate for β -galactosidase. The detection limit for enzyme on the plate reader was 15 fM of β Gal (Fig. 2). Second, we loaded the beads into femtoliter-volume well arrays and, after sealing a solution of RGP

into the wells of the array, allowed the signal from single enzymes to accumulate in the reaction chambers for 2.5 min, acquiring fluorescent images every 30 s. At the end of the experiment we used an image of the array that was acquired in white light to identify wells that contained beads (bead-containing wells scatter light differently than empty wells). The fluorescent images were used to determine which of those beads had an associated bound enzyme (from increasing intensity in time-lapsed fluorescent images). **Figure 2** shows a log-log plot of the percentage of beads that contained an enzyme as a function of bulk β Gal concentration. The lowest concentration of enzyme conjugate detected was 350 zeptomolar (zM) and the calculated limit of detection (LOD)—determined by extrapolating the enzyme concentration at a signal equal to background plus 3 s.d. of the background signal—was 220 zM. The sensitivity of SiMoAs to intrinsic label was, therefore, ~ 10 – 20 enzymes in 100 μl , corresponding to an increase in sensitivity over ensemble measurements using a typical ELISA-plate reader of a factor of about 68,000. For comparison, chemiluminescent detection of alkaline phosphatase⁴, a highly sensitive enzyme reporter system widely used in clinical diagnostics, has an LOD of about 30 aM (see <http://www.turnerbiosystems.com/doc/appnotes/S_0096.php>). The high sensitivity that arises from the thermodynamic and kinetic efficiency of the capture and detection processes in our approach are discussed below and detailed in **Supplementary Table 1**.

The linear dynamic range of digital detection of enzyme labels by SiMoAs is determined by the ability to distinguish "on" and "off" wells. At ratios of enzyme to beads of less than $\sim 1:10$, Poisson statistics show that the only statistically significant populations of beads are those carrying either one enzyme or no enzyme. Single enzymes can be detected provided that sufficient beads are interrogated and the number of active beads rises above the Poisson noise of counting active beads. At ratios of enzyme to beads greater than $\sim 1:10$, the fraction of active beads becomes much higher, and Poisson statistics show that there are a significant number of beads with multiple enzymes. To quantify the number of detected enzymes and maintain linearity in the subpopulations of beads with multiple enzymes, we use Poisson statistics to convert the number of active beads to the number of detected enzymes (see **Supplementary Table 1**). As the percentage of active beads approaches 50% (ratios of enzyme to beads greater than $\sim 1:1.5$), however, distinguishing "on" and "off" wells using image analysis software becomes challenging, and we reach a practical upper limit of the digital dynamic range. For example, the

Figure 2 Digitization of enzyme-linked complexes greatly increases sensitivity compared with bulk, ensemble measurements.



signal at 7 fM ($\sim 45\%$ active) in **Figure 2** deviates from linearity. As a result, the digital, linear dynamic range demonstrated here using 50,000 wells spanned approximately four orders of magnitude (from 3.5 fM down to 350 zM). Provided that proteins are labeled using appropriate enzyme concentrations (see below), this dynamic range is sufficient for many clinical applications.

We have developed digital ELISAs for two clinically relevant proteins—PSA and tumor necrosis factor- α (TNF- α)—to determine the sensitivity of the approach for detecting proteins in blood. The critical parameters in developing these assays were the concentrations of the beads and the two labeling reagents (detection antibody and enzyme conjugate). The choice of bead concentration depends on several competing factors. First, a sufficient number of beads must be present to capture most of the target analyte from thermodynamic and kinetic perspectives. Thermodynamically, 200,000 beads in 100 μ l, each of which has $\sim 80,000$ antibodies²⁵ bound to it, equates to an antibody concentration of about 0.3 nM. The antibody-protein equilibrium at that concentration permits a high capture efficiency ($>70\%$) (D.M.R., E.P.F., D.C.D., unpublished work). Kinetically, for 200,000 beads dispersed in 100 μ l, the average distance between beads is about 80 μ m. Proteins the size of TNF- α and PSA (17.3 and 30 kDa, respectively) will diffuse 80 μ m in <1 min, suggesting that capture of the protein molecules will not be limited kinetically over a 2-h incubation. Second, a sufficient number of beads must be present to be loaded onto the arrays to limit Poisson noise. Loading 200,000 beads into 50,000-well arrays typically results in 20,000–30,000 beads being trapped in femtoliter-sized wells. For a typical background signal of

1% active beads (see below), this loading results in a background signal of 200–300 active beads detected, corresponding to an acceptable coefficient of variation (CV) from Poisson noise of 6–7%. Third, excessive bead concentrations can lead to both increases in non-specific binding that reduces signal-to-background and low ratios of analyte to beads that can result in high CVs from Poisson noise. The balance of these factors means that 200,000 to 1,000,000 beads per 100 μ l of test sample is optimal for digital ELISA. The concentrations of detection antibody and enzyme conjugate were also minimized to yield the acceptable background signal (1%) and Poisson noise (**Supplementary Discussion**).

Figure 3 shows data from digital ELISAs for PSA and TNF- α . The human forms of the proteins were spiked into 25% bovine serum to final concentrations representative of clinical test samples. A fourfold dilution factor is typically used to reduce matrix effects in immunoassays⁴. Using digital ELISA to detect PSA in 25% serum, we obtained an LOD of ~ 50 aM (1.5 fg/ml), which equates to an LOD in whole serum of ~ 200 aM (6 fg/ml). The lowest concentration tested and detected was 250 aM in 25% serum, corresponding to 1 fM in whole serum. As LOD is determined by extrapolating the concentration at background plus 3 s.d. of the background, LODs for different runs are dependent on the CV of the background. We obtained subfemtomolar LODs of PSA in whole serum over several experiments with typical background variances. For comparison, a leading commercial PSA assay (ADVIA Centaur, Siemens) reports an LOD of 3 pM (0.1 ng/ml) in human serum, and ultrasensitive assays have been reported with LODs in the range

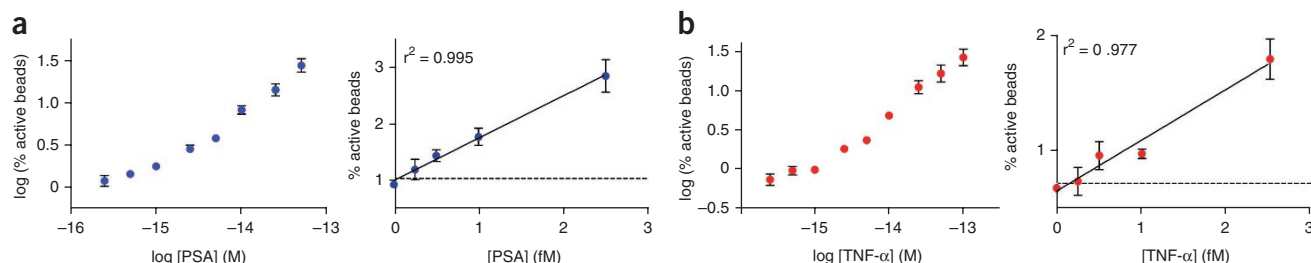
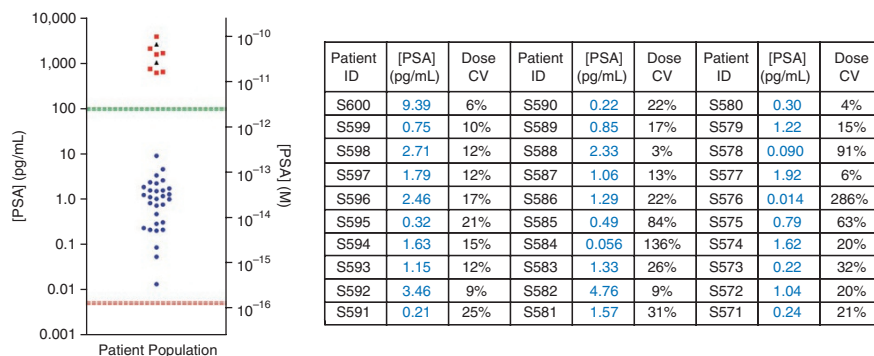


Figure 3 Subfemtomolar detection of proteins in serum using digital ELISA. (a,b) Changes in the percentage of active beads with changes in analyte concentration for human prostate-specific antigen (PSA) spiked into 25% serum (a) and human tumor necrosis factor- α (TNF- α) spiked into 25% serum (b). The concentrations plotted on the x axes refer to the final concentration of spiked protein in the diluted sample. The plots on the left-hand side show the assay response over the concentration range tested in log-log space. The plots on the right-hand side show the assay response in the femtomolar range in linear-linear space to illustrate the limit of detection (LODs) and linearity of response. LODs were determined by extrapolating the concentration from the signal equal to background signal plus 3 s.d. of the background signal. Broken lines, signal at the LOD. Error bars, s.d. over three replicates.

LETTERS

Figure 4 Digital detection of prostate-specific antigen (PSA) in serum samples of patients who had undergone radical prostatectomy. Concentrations of PSA in serum samples from radical prostatectomy patients (●), healthy control samples (■) and Bio-Rad PSA control samples (▲) determined using digital ELISA. Radical prostatectomy patient samples (SeraCare Life Sciences) all had undetectable PSA levels as measured by a leading clinical diagnostic assay (ADVIA Centaur); the green broken line represents the detection limit of the ADVIA Centaur PSA assay (100 pg/ml or 3 pM). All 30 patient samples were above the detection limit of the PSA digital ELISA, shown by the red broken line (0.006 pg/ml or ~200 aM), with the lowest patient PSA concentrations measured at 0.014 pg/ml (~400 aM) using digital ELISA. Patient samples with the lowest PSA levels approached the LOD of the assay, resulting in a large imprecision in the concentration determined (high dose %CV). The digital ELISA was validated for specificity to PSA using control standards (Bio-Rad) and serum from healthy individuals (ProMedDx) that had been assayed using the ADVIA Centaur PSA assay (**Supplementary Table 3**).



of 10–30 fM^{17,26}. The detection limit determined from the TNF- α digital ELISA was ~150 aM (2.5 fg/ml), corresponding to ~600 aM (10 fg/ml) in whole serum; the highest sensitivity commercially available ELISA for TNF- α has an LOD of 21 fM (0.34 pg/ml) in serum (**Supplementary Fig. 3**). The zero concentration spike of target protein for both assays provides a useful negative control for these experiments: 25% serum contains millimolar concentrations of a wide variety of proteins, and very low concentrations of target protein can be detected above these high-protein backgrounds. We also developed a proof-of-principle digital assay for DNA to show that low concentrations of nucleic acids can be detected without replication of the target (**Supplementary Fig. 2**).

The ability of digital ELISA to measure much lower concentrations of proteins than conventional ELISA derives from two effects: (i) the high sensitivity of SiMoAs to enzyme label and (ii) the low background signals that can be achieved by digitizing the detection of proteins. The sensitivity of any immunoassay is determined by the sensitivity of the detection technology to the label, the antibody affinity, the assay background and the variance (%CV) of the background measurement²⁷. The subattomolar sensitivity of SiMoAs to enzyme labels (**Fig. 2**) enables digital ELISA to detect subfemtomolar concentrations of labeled proteins. That said, for antibodies of given affinity, the sensitivity of the immunoassay will be determined by the assay background, and the high label sensitivity of SiMoAs helps reduce this background. Control experiments have shown that backgrounds in digital ELISA arise from nonspecific binding of detection antibody and enzyme conjugate to the capture bead surface (**Supplementary Table 2**). As SiMoAs provide superior label sensitivity over conventional assays, substantially less detection antibody (~1 nM) and enzyme conjugate (1–50 pM) are needed to detect binding events as compared with conventional assays (labeling reagent concentrations ~10 nM). The decreased label concentration reduces nonspecific binding to the capture surface, resulting in much lower background signals. For example, in the TNF- α and PSA digital ELISAs, the levels of nonspecific binding were equivalent to the signal produced by 1.8 fM and 1.2 fM of target protein, respectively (**Fig. 3**). The highest sensitivity commercial TNF- α assay has a level of nonspecific binding equivalent to 85 fM of TNF- α (**Supplementary Fig. 3**), which is 50 times more than we observe using digital ELISA. The ability to reduce backgrounds in digital ELISA by lowering the concentration of labeling reagents enables more sensitive immunoassays than are possible using conventional assays.

To demonstrate the possible diagnostic value of detecting very low concentrations of proteins in blood using digital ELISA, we measured PSA in serum samples from patients who had undergone radical prostatectomy. Levels of serum PSA are used both to screen for prostate cancer and to monitor recurrence of the disease in patients who have undergone radical prostatectomy²⁸. After radical prostatectomy the vast majority of PSA is eliminated, and concentrations fall below the detection limit of standard commercial assays (3 pM or 0.1 ng/ml). Although regular monitoring of these patients for increases in PSA concentrations can detect recurrence of prostate cancer, several years may pass after surgery for biochemical recurrence to be detected by current immunoanalyzers. The ability to accurately quantify PSA levels in patients who have undergone prostatectomy at very low concentrations (<300 fM or 10 pg/ml) should provide early indication of recurrence if PSA levels increase^{17,29}. **Figure 4** shows PSA concentrations measured using digital ELISA in the serum of 30 patients (age 60–89) who had undergone radical prostatectomy and whose blood was collected at least 6 weeks after surgery. The PSA concentrations in the sera of all 30 patients were below the detection limit of commercial assays. Whole-serum samples were diluted 1:4 in buffer and measured using the digital ELISA specific for PSA (**Fig. 3a**). PSA was successfully detected in all 30 patients using digital ELISA, with concentrations ranging from 14 fg/ml to 9.4 pg/ml, with an average of 1.5 pg/ml. Further clinical studies are required to establish the diagnostic benefit of measuring PSA at femtogram per milliliter amounts in patients following radical prostatectomy.

By isolating and detecting single immunocomplexes in arrays of femtoliter-volume wells, digital ELISA enables clinically important proteins in serum to be measured at subfemtomolar concentrations. We believe that the improvement in sensitivity shown by digital ELISA over previously reported approaches will translate into diagnostic benefits. For example, PSA concentrations in 9 of the 30 radical prostatectomy samples assayed in this work fell below the LOD of the previously highest sensitivity PSA assay based on nanoparticle labels¹⁷. An attractive feature of this approach is the ability to directly use reagents developed for standard ELISA for substantially more sensitive assays. We continue to improve the SiMoA technology in two key areas. First, based on the sensitivity to enzyme label (**Fig. 2**), it seems that the sensitivity of protein detection could be increased at least 100-fold if nonspecific interactions that cause background signals could be minimized. The ability to isolate and interrogate single molecules on individual beads provides avenues for distinguishing antibody-antigen

binding events from nonspecifically bound complexes. Second, we are simplifying the logistics of the assays. Even in its present form, however, we believe that digital ELISA has the potential to facilitate earlier diagnosis and treatment of disease.

METHODS

Methods and any associated references are available in the online version of the paper at <http://www.nature.com/naturebiotechnology/>.

Note: Supplementary information is available on the Nature Biotechnology website.

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AUTHOR CONTRIBUTIONS

D.M.R., C.W.K., D.R.F., D.R.W. and D.C.D. conceived the approach. D.R.F. built the imaging system. D.M.R., C.W.K., T.G.C., S.C.H., L.S., P.P.P., A.J.R., E.P.F., J.D.R. and G.K.P. conducted the experiments. T.P. wrote the image analysis software. L.C. prepared reagents. D.M.R. and D.C.D. wrote the manuscript. All authors were involved in designing experiments, reviewing and discussing data, and commenting on the manuscript.

COMPETING FINANCIAL INTERESTS

The authors declare competing financial interests: details accompany the full-text HTML version of the paper at <http://www.nature.com/naturebiotechnology/>.

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ONLINE METHODS

Materials. Optical fiber bundles were purchased from Schott North America. Nonreinforced gloss silicone sheeting was obtained from Specialty Manufacturing. Hydrochloric acid, anhydrous ethanol and molecular biology-grade Tween-20 were all from Sigma-Aldrich. Carboxyl-terminated magnetic beads (2.7- μm diameter) were purchased from Varian. Monoclonal capture antibody to human TNF- α , polyclonal detection antibody to human TNF- α and recombinant human TNF- α were purchased from R&D Systems. Monoclonal capture antibody to PSA, monoclonal detection antibody to PSA and purified PSA were purchased from BiosPacific; the detection antibody was biotinylated using standard methods. 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride, N-hydroxysulfosuccinimide and SuperBlock T-20 Blocking Buffer were purchased from Thermo Scientific. Purified DNA was purchased from Integrated DNA Technologies. S β G was conjugated in the laboratory using standard protocols. RGP was purchased from Invitrogen. The fiber polisher and polishing consumables were purchased from Allied High Tech Products.

Preparation of magnetic beads presenting biotin-labeled DNA and capture of enzyme conjugate (Fig. 2). Beads functionalized with DNA capture probe (5'-NH₂/C12-GTT GTC AAG ATG CTA CCG TTC AGA G-3') were prepared according to the manufacturer's instructions. These beads were incubated with 1 μM of biotinylated complementary DNA (5'-biotin-C TCT GAA CGG TAG CAT CTT GAC AAC-3') overnight (16 h) in TE buffer containing 0.5 M NaCl and 0.01% Tween-20. After incubation the beads were washed three times in PBS buffer containing 0.1% Tween-20. The bead stock was distributed into a microtiter plate giving 400,000 beads per well in 100 μl . The buffer was aspirated from the microtiter plate wells; the beads were resuspended and incubated with various concentrations of S β G in Superblock containing 0.05% Tween-20 for 4 h. The beads were then separated and washed six times with 5 $\times$ PBS buffer containing 0.1% Tween-20. For detection of enzyme, the beads were either (i) resuspended in 20 μl of PBS containing 0.1% Tween-20, and 10- μl aliquots were loaded onto two femtoliter-volume well arrays for SiMoA detection; or (ii) resuspended in 100 μl of 100 μM RGP in PBS, incubated for 1 h at 23 $^{\circ}\text{C}$ and read on a fluorescence plate reader (Infinite M200, Tecan).

Capture of proteins on magnetic beads and formation of enzyme-labeled immunocomplex (Figs. 3 and 4). Beads functionalized with an antibody to the target protein were prepared according to the manufacturer's instructions. Test solutions (100 μl) containing the protein of interest were incubated with suspensions of 200,000 magnetic beads for 2 h at 23 $^{\circ}\text{C}$. The beads were then separated and washed three times in PBS and 0.1% Tween-20. The beads were resuspended and incubated with solutions containing detection antibody (typically ~ 1 nM) for 45 min at 23 $^{\circ}\text{C}$. The beads were then separated and washed three times in PBS and 0.1% Tween-20. The beads were incubated with solutions containing S β G (1–50 pM) for 30 min at 23 $^{\circ}\text{C}$, separated and washed six times in PBS and 0.1% Tween-20. The beads were then resuspended in 10 μl of PBS and loaded onto a femtoliter-volume well array. The total time of the assay was ~ 6 h.

Capture of DNA on magnetic beads and formation of enzyme-labeled complex (Supplementary Fig. 2). A total of 200,000 beads functionalized with DNA capture probe were incubated with 100- μl solutions containing the target DNA (5'-TT GAC GGC GAA GAC CTG GAT GTA TTG CTC C TCT GAA CGG TAG CAT CTT GAC AAC-3') for 2 h. After incubation, the DNA target solution was removed and the beads were washed three times in 0.2 $\times$ SSC buffer containing 0.1% Tween-20. The beads were then resuspended and mixed with 10 nM biotinylated signal probe (5'-TAC ATC CAG GTC TTC GCC GTC AA/Biotin/-3') that is also specific to the target DNA for 1 h. The beads were then washed three times in 0.2 $\times$ SSC buffer containing 0.1% Tween-20 after removing the signal probe. A solution of 10 pM S β G was then added to the bead pellet, resuspended and mixed for 1 h. The beads were separated and washed six times in 5 $\times$ PBS buffer containing 0.1% Tween-20. The beads were then resuspended in 10 μl of PBS and loaded onto a femtoliter-volume well array.

Preparation of femtoliter-volume well arrays. Optical fiber bundles approximately 5 cm long were sequentially polished on a polishing machine using 30-, 9- and 1- μm -sized diamond lapping films. The polished fiber bundles were chemically etched in a 0.025 M HCl solution for 130 s and then immediately submerged into water to quench the reaction. The etched fibers were sonicated for 5 s in water, washed in water for 5 min and dried under vacuum. The differential etch rate of the core and cladding glass of the fiber bundle arrays caused 4.5- μm -diameter wells to be formed in the core fibers³⁰. Different etch times resulting in different well depths were initially investigated. If wells were too deep, then multiple beads were deposited in each well and sealing was disrupted; if wells were too shallow, then the beads were not retained in the wells and poor loading efficiencies were observed. Well depths of 3.25 ± 0.5 μm were optimal for retaining single beads in wells while maintaining good seals.

Loading of beads into femtoliter-volume well arrays. A short length of PVC tubing was placed on the etched end of a fiber bundle to create a reservoir to hold the bead solution. Ten microliters of the concentrated bead solution were pipetted into this reservoir. The fiber bundle was then centrifuged at 1,300g for 10 min to force the beads into the etched wells. The PVC tubing was removed after centrifugation. The fiber bundle was dipped in PBS solution to wash off excess bead solution, and the surface was swabbed with deionized water. In addition to well depth (see above), bead concentration was an important parameter for maximizing bead loading efficiencies. Above concentrations of 200,000 beads per 10 μl loaded, typically 40–60% of wells in a 50,000-well array were occupied by a single bead, resulting in percentage active beads with acceptable Poisson noise. At concentrations below 200,000 beads per 10 μl loaded, bead loading efficiency dropped, resulting in fewer active beads and higher Poisson noise. In these experiments, therefore, at least 200,000 beads per reaction were used and loaded onto the arrays.

Detection of beads and enzyme-labeled beads in femtoliter-volume well arrays. A custom-built imaging system containing a mercury light source, filter cubes, objectives and a CCD camera was used for acquiring fluorescence images^{21,23}. Fiber bundles were mounted on the microscope stage using a custom fixture. A droplet of β -galactosidase substrate (RGP) was placed on the silicone gasket material and placed in contact with the well arrays. A precision mechanical platform moved the silicone sheet into contact with the end of the fiber bundle, creating an array of isolated femtoliter-volume reaction vessels. Fluorescence images were acquired (558 nm excitation; 577 nm emission) with an exposure time of 1,011 ms. Five frames (at 30-s intervals) were taken for each femtoliter-volume well array. The product of the enzymatic reaction used in these studies—resorufin—has high photostability with a low photobleaching rate (rate of photobleaching, $k_{\text{ph}} = 0.0013 \text{ s}^{-1}$)²¹, making multiple exposures possible. We performed time-course fluorescence measurements (i) to allow stable fluorescent artifacts to be removed from images and (ii) to ensure that the signal from a beaded well was from an enzyme. For (i), the first fluorescent image was subtracted from fluorescent images acquired at each subsequent time point. This process removed light intensity that did not change with time; for example, fluorescence from dust and scattered light. For (ii), a positive or “on” well was identified only where fluorescence intensity in a beaded well increased in every frame and by at least 20% over four frames. This process removed false positives from random changes in fluorescence during image acquisition. **Supplementary Figure 1** shows histograms of fluorescence from wells with and without enzymes, showing the good distinction between “on” and “off” wells. Arrays were also imaged with white light to identify those wells that contain beads. After acquiring the fluorescence images, the arrays were illuminated with white light and imaged on the CCD camera. Due to scattering of light by the beads, those wells that contained a bead appeared brighter in the image than wells without beads. The fluorescence and white-light images were analyzed using customized software.

30. Pantano, P. & Walt, D.R. Ordered nanowell arrays. *Chem. Mater.* **8**, 2832–2835 (1996).